

Molecular signaling in muscle is affected by the specificity of resistance exercise protocol

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Mammalian target of rapamycin and mitogen-activated protein kinase (MAPK) signaling pathways have been highlighted as important for muscle adaptations and thus, they may distinguish adaptations to different exercises. Typically, resistance exercise designed for muscle hypertrophy has moderate intensity (60–80% of one repetition maximum, 1 RM) while one prioritizing maximal strength with minor hypertrophy has a higher intensity ($\geq 90\%$ of 1 RM). Eight untrained men (28.4 ± 3.7 years) conducted two different bilateral leg press exercise protocols: hypertrophic (5×10 RM) and pure maximal strength (15×1 RM) in a counterbalanced, cross-over design with 1 week between exercises. Vastus lateralis muscle biopsies were taken before

and 0.5 h after resistance exercise, or in six controls (26.5 ± 3.6 years) who rested. The phosphorylation of p70S6K (Thr⁴²¹/Ser⁴²⁴), rpS6 (Ser^{240/244} and Ser^{235/236}) and MAPK p38 as increased (~ 2 –16 fold) after both exercise protocols. However, the phosphorylation of MAPK Erk1/2 and p70S6K at Thr³⁸⁹ increased only after 5×10 RM. The increase in the phosphorylation of p70S6K (Thr⁴²¹/Ser⁴²⁴), rpS6 (Ser^{235/236}) and Erk1/2 were higher after 5×10 RM ($P < 0.05$). No changes were seen in controls. In conclusion, MAPK signaling is greater after hypertrophic than maximal strength exercise protocol. This may mediate adaptations specific to these different types of training regimens.

Resistance training-induced increase in muscle strength is achieved through either muscle size or neural adaptations or both. The prevailing opinion is that resistance exercise with a high-intensity and a low amount of repetitions per set, primarily designed to maximize neural adaptation, does not normally lead to large changes in muscle hypertrophy (Häkkinen et al., 1987; Fry 2004). Thus, “hypertrophic” training traditionally consists of a rather large total number of repetitions (~ 8 –12) per set with medium loads [60–85% of one repetition maximum (RM)] (Tesch & Larsson, 1982; Tesch, Colliander, & Kaiser, 1986). This is in contrast to a “maximal strength” type of training among weightlifters or powerlifters (90–100% loads of 1 RM, one to three repetitions per set), who aim to increase muscle strength in their weight-class primarily through neural adaptations (Häkkinen, Alen, & Komi, 1985; Häkkinen et al., 1987). Moreover, resistance training with high repetitions can enhance high-intensity endurance capacity locally in muscle in nonendurance trained athletes when compared with maximal strength type of training (Campos et al., 2002).

Although we may predict the expected outcomes of these diverse types of resistance training modes, we do not know the exact underlying physiological mechanisms that govern these adaptations in human muscle. Kumar et al. (2009) showed recently that resistance exercise of close to maximal strength type of exercise six sets of three repetitions (90% of 1 RM), produced comparable muscle myofibrillar protein synthesis 1–2-h post-exercise compared with an equal total volume load bouts with three sets of eight to nine repetitions (60–75% of 1 RM). In that study, due to large variation and a low subject number, the mammalian target of rapamycin (mTOR) signaling responses to different intensities could not be, according to the authors, thoroughly compared. Previously in our laboratory, a pure maximal strength exercise (20×1 RM) induced only small systemically measured metabolic changes (e.g., blood lactate and hormones) compared with a hypertrophic protocol (Häkkinen & Pakarinen, 1993). However, the effects in skeletal muscle of pure maximal strength exercise protocol with only one repetition per set remain to be investigated.

Muscle hypertrophy due to resistance training appears largely to result from cumulative acute increases in muscle (myofibrillar) protein synthesis (Wilkinson et al., 2008). Phosphorylation of S6K1 (p70 and p85 ribosomal S6 kinases), most probably through mTOR, has been shown to be important in muscle protein synthesis (Kumar et al., 2009) and hypertrophy (Baar & Esser, 1999; Terzis et al., 2008). In addition, mitogen-activated protein kinase (MAPK) family proteins Erk1/2 and p38 isoforms can also positively regulate muscle size (Haddad & Adams, 2004; Shi et al., 2009), but they may also be involved in muscle endurance adaptations (Nader & Esser, 2001; Higginson et al., 2002; Zhou et al., 2007; Pogozelski et al., 2009).

It is not known, whether a pure form of maximal strength exercise protocol with large but short duration mechanical stimuli induce similar early signaling responses in skeletal muscle compared with a hypertrophic protocol associated with considerable array of metabolic/ionic changes, which *per se* may augment some signaling pathways. Thus, the purpose of this randomized trial was to examine the signaling pathways through which the muscle possibly distinguishes these specific types of resistance exercise stimulus. We hypothesized that the present maximal strength type of exercise would produce only limited response compared with the hypertrophic protocol.

Materials and methods

Subjects

Eight young men ($n = 8$, age 28.4 ± 3.7 years) conducted two different resistance exercise protocols with a balanced randomized order. Six additional men (26.5 ± 3.6 years) served as controls. Anthropometric details of the subjects are presented in Table 1. The subjects were recruited for the study by advertising in newspapers and through email lists.

All the subjects were examined by a physician and none had medical problems that might confound the results of this investigation. None of the subjects had any experience in regular resistance training but they were recreationally active. Their normal habitual activities included walking, jogging, swimming or ball-games and they were urged to continue their normal activities and lifestyle exactly in the same manner during the experimental period. Before the investigation, each subject was informed about the experimental design and possible associated risks and discomforts. Each subject then signed informed consent to participate in the study. The study

was approved by the local Ethics Committee of the University of Jyväskylä and was conducted in accordance with the Declaration of Helsinki.

Design

This investigation examined acute responses to two different loading protocols designed specifically for either maximal strength (i.e. neural) or hypertrophic adaptations. The subjects conducted "hypertrophic" 5×10 RM (5×10 RM) and pure "maximal strength" 15×1 RM (15×1 RM) exercise protocols in a balanced randomized order in a cross-over design and with 7 days between the loadings (Fig. 1). An additional group of control subjects just rested without exercising. All the measurements were always carried out at the same time of day (average 15–18:00 hours) to exclude the effects of daily variations (Sedliak et al., 2007). All the measurements were preceded by at least 2 days of rest from physical activity.

Dietary intake

We wanted to avoid a fasting state as it is not a normal situation to conduct heavy resistance exercise, and thus, a standardized macronutrient drink was ingested 3 h before the start of the each protocol [0.5 g powder in water per 1 kg body mass, consisting of 68% carbohydrate, 30% milk protein, 2% fat (Skip Ltd, Stockholm, Sweden)]. Furthermore, as hydration status affects strength performance (Judelson et al., 2007) and physiological responses to exercise (Judelson et al., 2008), the subjects ingested 0.5 liters of water 1 h pre-exercise.

Dietary intake was recorded in diaries for 2 days before the first biopsy day at the start of the study and on the biopsy day. The subjects were instructed to replicate their diet as closely as possible before the second loading day.

Heavy resistance exercise protocols

A bilateral leg press machine (David 210, David Sports Ltd, Helsinki, Finland) was used for the heavy resistance exercise. The hypertrophic protocol was similar to our earlier studies using the 5×10 RM protocol (Häkkinen & Pakarinen, 1993; Hulmi et al., 2007; Hulmi et al., 2009a,b). The 15×1 RM protocol was close to a previous 20×1 RM protocol used in our laboratory (Häkkinen & Pakarinen, 1993) and can be considered as close to a "pure" maximal strength protocol as possible. The first set in the 5×10 RM protocol started with the 80% 1 RM load based on the earlier strength test day to measure baseline strength of the subjects and to familiarize the subjects to the exercises. The loads were adjusted during the course of the resistance exercise bout due to fatigue so that each subject would be able to perform 10 repetitions at each set. In the 15×1 RM protocol, the maximal load successfully lifted in the familiarization session was used as the initial load (set 1). Again, the load was adjusted throughout the exercise due to fatigue so that the subject could perform only one repetition with the proper technique. The range of motion was from 60° knee angle position to the full-extended position. If the load was too heavy, the subject was assisted slightly during the last repetition(s) of the set. The recovery between the sets was 3 min in 15×1 RM and 2 min in 5×10 RM loading protocols.

Maximal force

The David 210 leg press (David Sports Ltd) was used to measure maximal bilateral concentric force production for leg extensors (hip and knee extensors) during a familiarization

Table 1. Subject characteristics

	Age (years)	Height (cm)	Mass (kg)	Fat (%)
Resistance exercise				
Subjects ($n = 8$)				
Mean	28.4	180.3	78.7	15.4
SD	3.7	3.9	10.2	3.3
Controls ($n = 6$)				
Mean	26.5	180.5	74.9	17.4
SD	3.6	4.7	12.6	4.8

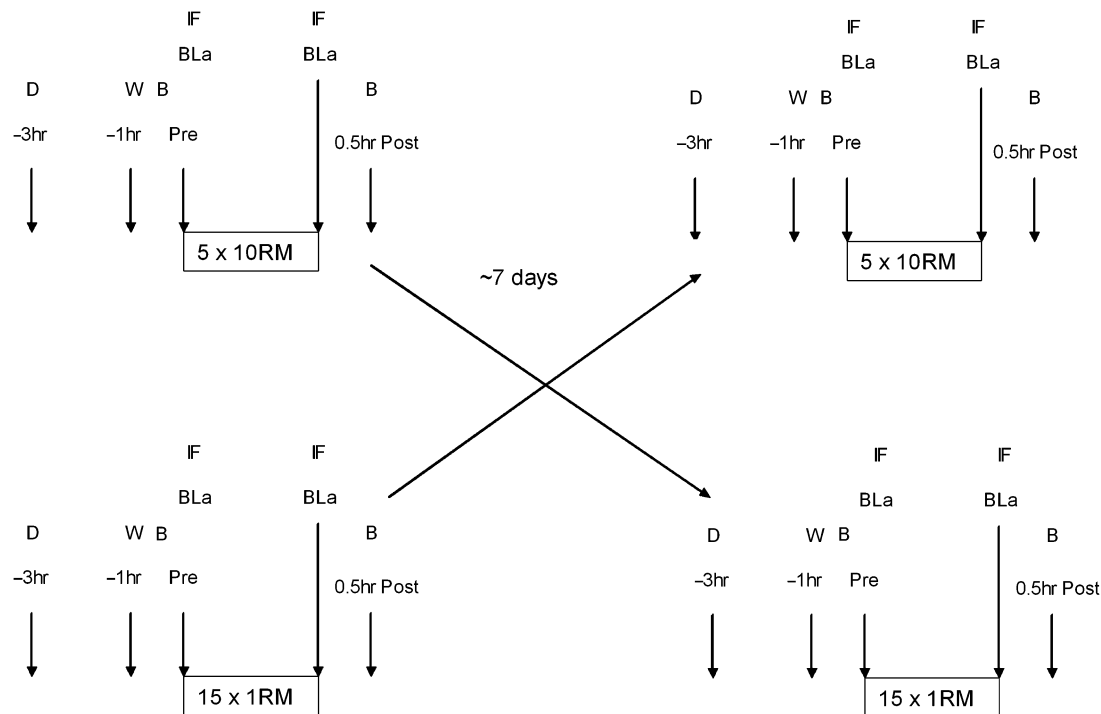


Fig. 1. Experimental design for the balanced, cross-over resistance exercise protocols. B, vastus lateralis muscle biopsy; BLa, blood lactate sampling; D, standardized macronutrient drink (see text); IF, isometric force of leg extensors; W, water ingestion. Resistance exercise was either 5 × 10 RM (hypertrophic) or 15 × 1 RM (maximal strength) leg press. Additional control-subjects just rested the similar time-period without conducting exercise. 1 RM, one repetition maximum.

session. Separate trials were performed for concentric 1 RM testing during a familiarization session before the loadings. After each repetition, the load was increased until the subject was unable to extend his legs to the full-extended position. The highest successfully lifted load was determined as the 1 RM.

Maximal bilateral isometric force of the leg extensor muscles was measured before and after the resistance exercise protocols on an electromechanical dynamometer with a knee angle of 107° and hip angle of 110° (Häkkinen et al., 2001; Hulmi et al., 2009a). A minimum of two trials were taken before exercise and only one trial after exercise to enable blood lactate to be taken within 1 min of the completion of the last set. The force signal of the isometric measurements was recorded and analyzed with Signal software version 2.16 (Cambridge Electronic Design Ltd, Cambridge, UK). The devices used in the present study have high reliability for measuring the isometric force and concentric 1 RM (Hulmi et al., 2009a). The subjects were carefully familiarized with the test procedures and had several warm-up contractions in all devices before testing.

Anthropometry

During the familiarization session, height, body mass (kg) and fat percentage were measured (InBody 720 body composition analyzer, Biospace Co. Ltd, Seoul, South Korea).

Muscle biopsies

Muscle biopsies were obtained 0.5 h before and 0.5 h after the bout of resistance exercise or before and after about similar time of resting (1.5 h) in the control group (Fig. 1). Biopsies were taken from the vastus lateralis muscle with a 5 mm

Bergström biopsy needle and with suction, midway between the patella and greater trochanter. The pre-exercise and pre-controlled resting period biopsy was always from the right leg and post-exercise or post-resting biopsy from the left leg. To avoid any residual effects of the previous biopsies, the biopsies in the second bout were obtained ~ 1 cm above the previous biopsy location.

The muscle sample was cleaned of any visible connective and adipose tissue as well as blood and frozen immediately in liquid nitrogen (− 180 °C) and stored at − 80 °C.

Tissue processing

Muscle biopsy specimens were hand-homogenized in ice-cold buffer (Hulmi et al., 2009b) (20 mM HEPES (pH 7.4), 1 mM EDTA, 5 mM EGTA, 10 mM MgCl₂, 100 mM β-glycerophosphate, 1 mM Na₃VO₄, 2 mM DTT, 1% Triton X-100, 0.2% sodium deoxycholate, 30 μg/mL leupeptin, 30 μg/mL aprotinin, 60 μg/mL PMSF, and 1% phosphatase inhibitor cocktail [P 2850; Sigma, St Louis, Missouri, USA]) at a dilution of 15 μL/mg of wet weight muscle. Homogenates were rotated for 30 min at 4 °C, centrifuged at 10 000 g for 10 min at 4 °C to remove cell debris, and stored at − 80 °C. Total protein content was determined using the bicinchoninic acid protein assay (Pierce Biotechnology, Rockford, Illinois, USA).

Western immunoblot analyses

Aliquots of muscle lysate were solubilized in Laemmli sample buffer and heated at 95 °C to denature proteins. Samples containing 30 μg of total protein were separated by SDS-PAGE for 90 min at 200 V using 4–20% gradient gels on Criterion electrophoresis cell (Bio-Rad Laboratories,

Richmond, California, USA). All four samples from each subject were run on the same gel. Proteins were transferred to PVDF membranes at 350 mA constant current for 3 h on ice at 4 °C. Membranes were blocked in TBS with 0.1% Tween 20 (TBS-T) containing 5% nonfat dry milk for 1 h and then incubated overnight at 4 °C with rabbit polyclonal primary antibodies. Antibodies recognized phosphorylated mTOR at both Ser²⁴⁴⁸ and at Ser²⁴⁸¹ (Copp, Manning, & Hunter, 2009), rpS6 at both Ser^{235/236} and at Ser^{240/244} as the former appears to be more specific to MAPK pathway (Pende et al., 2004; Roux et al., 2007) and the latter probably more universal or specific to the mTOR pathway (Pende et al., 2004; Roux et al., 2007). Moreover, antibodies were against p38 MAPK at Thr¹⁸⁰/Tyr¹⁸², and Erk1/2 (p44/p42) at Thr²⁰²/Tyr²⁰⁴ and p70^{S6K} at both Thr³⁸⁹ and Thr⁴²¹/Ser⁴²⁴ (Cell Signaling Technology, Beverly, Massachusetts, USA). Moreover, total proteins of p70^{S6K} and p85^{S6K} (rabbit polyclonal) (Santa Cruz Biotechnology, Santa Cruz, California, USA) and rpS6 (rabbit monoclonal) (Cell Signaling Technology) with respective antibodies were analyzed to verify that the total protein content of part of the phosphospecific proteins analyzed of signaling proteins does not significantly change during this short time period of 0.5 h post-exercise. The uniformity of protein loading was confirmed by staining the membrane with Ponceau S and by re-probing the membrane with an antibody against α -actin (Sigma).

All the primary antibodies were diluted 1:2000 in TBS-T containing 2.5% nonfat dry milk except against α -actin, which was diluted 1:20 000. Membranes were then washed in TBS-T, incubated with secondary antibody (horseradish peroxidase-conjugated anti-rabbit IgG; Cell Signaling Technology) diluted 1:25 000 in TBS-T with 2.5% milk for 1 h followed by washing in TBS-T. Phosphorylated proteins were visualized by ECL according to the manufacturer's protocol (Super-Signal west femto maximum sensitivity substrate, Pierce Biotechnology) and quantified (band intensity $\times$ volume) using a ChemiDoc XRS in combination with Quantity One software (version 4.6.3. Bio-Rad Laboratories).

Blood lactate

Fingertip blood lactate samples were collected immediately before and after exercises into capillary tubes, which were placed in a 1 mL hemolyzing solution and analyzed automatically (EKF diagnostic, Biosen, Barleben, Germany).

Statistical analyses

All data are expressed as means $\pm$ SD, except where designated. The groups/conditions were analyzed by a two-way analysis of variance (ANOVA). When differences between the groups were found, the changes from pre to post between the 5 $\times$ 10 RM, 15 $\times$ 1 RM and controls (changes) were analyzed by a one-way ANOVA and as a *post hoc* test as this was a cross-over design, a dependent *t*-test was used when comparing two resistance exercise conditions or independent *t*-test when comparing exercise conditions with the separated control group. SPSS version 15.0 for Windows (SPSS Inc., Chicago, Illinois, USA) and Microsoft Excel for Windows XP were used for statistical analyses. The level of significance was set at $P < 0.05$.

Results

Resistance exercise bouts

The total volume of the work performed in the resistance exercise bout (loads $\times$ sets $\times$ repetitions) was significantly higher ($P < 0.05$) during the

5 $\times$ 10 RM (7 680 $\pm$ 1 148 kg) compared with the 15 $\times$ 1 RM (2 670 $\pm$ 403 kg). Blood lactate increased significantly pre to post-exercise during both exercise conditions ($P < 0.01$), but the increase was significantly greater due to 5 $\times$ 10 RM (post-exercise 11.6 $\pm$ 2.3 mmol/L) compared with 15 $\times$ 1 RM (4.8 $\pm$ 2.4 mmol/L) ($P < 0.01$). Similarly, maximum isometric force decreased significantly post-exercise during both conditions ($P < 0.001$), but the decrease was greater after 5 $\times$ 10 RM ($-45.4 \pm 10.5\%$) protocol than after 15 $\times$ 1 RM ($-21.4 \pm 6.6\%$) ($P < 0.01$).

Western blotting results

The phosphorylation of p70^{S6K} at Ser⁴²⁴/Thr⁴²¹, rpS6 at Ser^{235/236} and at Ser^{240/244}, and MAPK p38 at Thr¹⁸⁰/Tyr¹⁸² were increased after both exercise protocols and also when compared with controls ($P < 0.05$) (Fig. 2 and Fig. 3). The phosphorylation of p70^{S6K} at Thr³⁸⁹ increased only after 5 $\times$ 10 RM ($P < 0.01$) but there was a trend for an increase after 15 $\times$ 1 RM also ($P = 0.10$) (Fig. 2). The phosphorylation of mTOR at Ser²⁴⁴⁸ and at Ser²⁴⁸¹ remained statistically unchanged (Fig. 2).

The increase in the phosphorylation of p70^{S6K} at Ser⁴²⁴/Thr⁴²¹ and rpS6 at Ser^{235/236} (Fig. 2) were higher after 5 $\times$ 10 RM ($P < 0.05$). The phosphorylation of Erk1/2 at Thr²⁰²/Tyr²⁰⁴ increased only after 5 $\times$ 10 RM and these changes were higher when compared with 15 $\times$ 1 RM (Fig. 3). The two pooled p38 bands showed that the difference with the protocols did not reach statistical significance ($P = 0.14$) (not shown). However, when the two clear bands of p38 were analyzed separately, the upper band, purportedly p38 γ (Boppart et al., 2000), showed a higher change in 5 $\times$ 10 RM (Fig. 3). The change in the phosphorylation of rpS6 at Ser^{240/244} and p70^{S6K} at Thr³⁸⁹ were higher after 5 $\times$ 10 RM than after 15 $\times$ 1 RM in 6/8 subjects but this did not reach statistical significance ($P > 0.3$) in contrast to the rpS6 at Ser^{235/236} and p70^{S6K} at Ser⁴²⁴/Thr⁴²¹ as mentioned earlier (Fig. 2). There was no change in the total protein expression of p70^{S6K}, p85^{S6K} and rpS6 (see blot images in Fig 3).

Discussion

The major findings in the present study were that (1) MAPK signaling is higher 0.5 h after the hypertrophic resistance exercise protocol (5 $\times$ 10 RM) compared with the maximal strength protocol (15 $\times$ 1 RM). However, (2) we showed for the first time that even the latter, with only one repetition per set and small peripheral/local metabolic changes can acutely increase MAPK signaling and to a smaller extent also signaling downstream of mTOR, in previously untrained young adult males in the fed state.

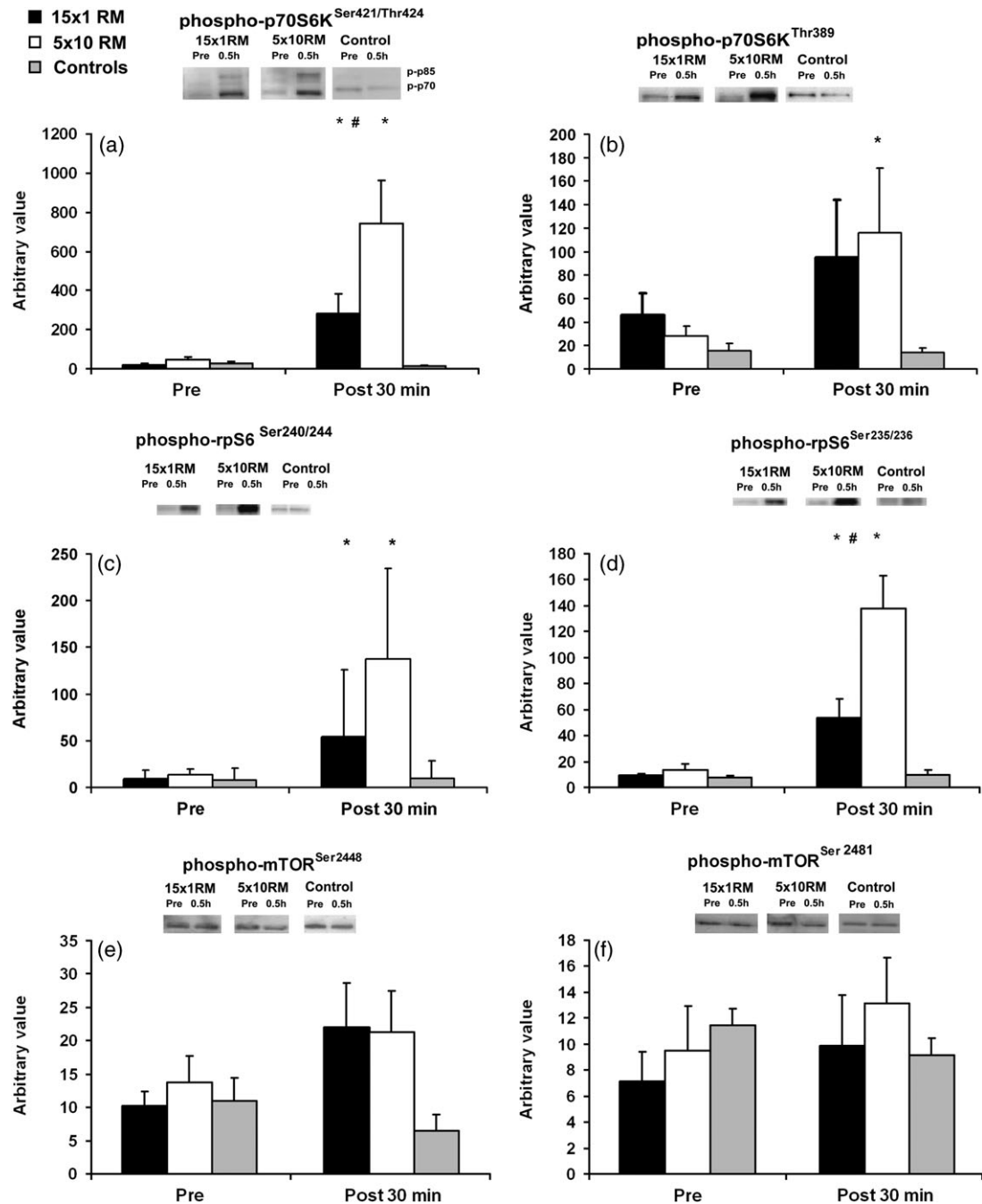


Fig. 2. Phosphorylation of (a) p70^{S6K} at Ser⁴²⁴/Thr⁴²¹, (b) p70^{S6K} at Thr³⁸⁹, (c) rpS6 at Ser^{240/244}, (d) rpS6 at Ser^{235/236}, (e) mTOR at Ser²⁴⁴⁸ and (f) mTOR at Ser²⁴⁸¹ before (pre) and 30 min after (post 30 min) either 5 × 10 RM protocol or 15 × 1 RM protocol or controlled resting (Controls). Immunoblot of one individual is shown on top of each figure. p70^{S6K} figure (blot for the p70^{S6K} at Ser⁴²⁴/Thr⁴²¹) shows also that phosphorylation of other isoform of S6K1, p85^{S6K} at Thr⁴⁴⁴/Ser⁴⁴⁷ followed always the same trend as p70^{S6K} band below. The bar below depicts only quantification of the p70^{S6K}. Values in the figures are arbitrary units (means ± SE). **P* < 0.05 vs pre, #*P* < 0.05 difference between 5 × 10 RM and 15 × 1 RM in the change from pre to post. *N* = 8 for 5 × 10 RM and 15 × 1 RM and *n* = 6 for the control group. 1 RM, one repetition maximum.

The phosphorylation of p38 increased after both types of resistance exercise protocols, but the Erk1/2 phosphorylation only increased after the hypertrophic protocol. Increased p38 and Erk1/2 phosphorylation has been previously reported after hypertrophic type of resistance exercise bout (Karls-son et al., 2004; Deldicque et al., 2008; Holm et al.,

2010). Interestingly, we found that only the larger sized band, purportedly representing the γ -isoform of p38 had a significantly higher increase in its phosphorylation state after 5 × 10 RM when compared with 15 × 1 RM. At least in mice, p38 γ isoform appears to be a key regulator in the metabolic adaptation of skeletal muscle (Pogozelski et al.,

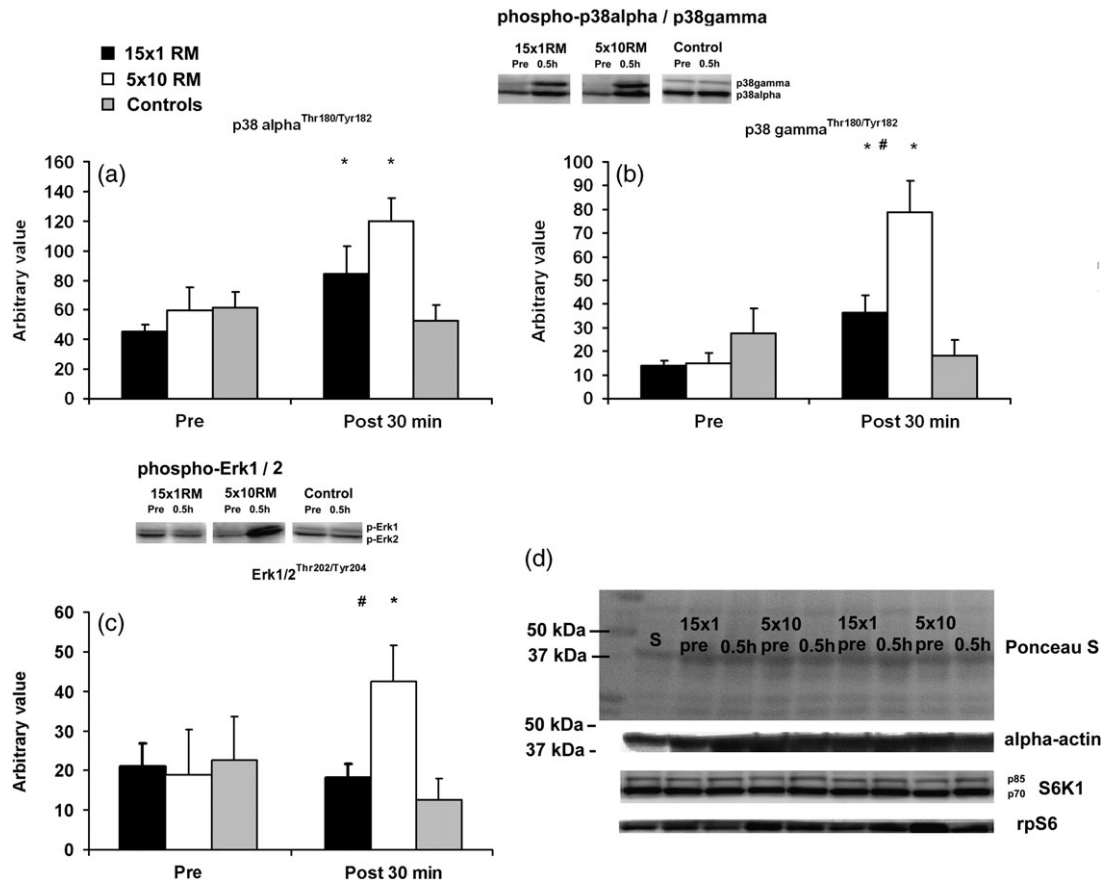


Fig. 3. Phosphorylation of (a) p38 α/β at Thr¹⁸⁰/Tyr¹⁸², (b) p38 γ at Thr¹⁸⁰/Tyr¹⁸², (c) Erk1/2 at Thr²⁰²/Tyr²⁰⁴ and (d) Ponceau S staining shows an equal protein loading in an example membrane in which lane 1 is a molecule marker (Precision Plus Protein™ Dual Color Standards, Biorad, Benicia, California, USA) and lane two pooled standard (15 μ g of total protein). Lanes 3–18 show alternating 5 $\times$ 10 RM and 15 $\times$ 1 RM conditions (30 μ g of total protein per lane). The strongest band at $\sim$ 42 kDa is probably largely due to abundant skeletal α -actin. Blotting against α -actin and total p70^{S6K}, p85^{S6K} and rpS6 are shown below ponceau S. The pooling of the two p38 bands showed similarly an increase after both exercise protocols than when analyzed separately, but contrastingly, the difference with the protocols did not reach statistical significance ($P = 0.14$) (not shown). * $P < 0.05$ vs pre, # $P < 0.05$ difference between 5 $\times$ 10 RM and 15 $\times$ 1 RM in the change from pre to post. $N = 8$ for 5 $\times$ 10 RM and 15 $\times$ 1 RM and $n = 6$ for the control group. 1 RM, one repetition maximum.

2009), and in humans, long-duration endurance running increased the phosphorylation of p38 γ , but no change was seen in the phosphorylation of p38 α/β (Boppart et al., 2000). Thus, it can be speculated that the phosphorylation of p38 γ may be responsive to metabolic/ionic perturbations in addition to mechanical tension, but the α/β isoform may respond relatively less to the metabolites and ions produced by anaerobic metabolism.

Typical hypertrophic resistance exercise protocols, and also in the present study, have usually larger total workload than in maximal strength protocols. At the same time, larger metabolic/ionic changes were observed after the hypertrophic protocol as can be predicted from the substantially larger fatigue (acute decrease in maximal isometric force production), and larger blood lactate responses, both typically coinciding with, e.g., acidosis (Robergs Ghiasvand & Parker, 2004). Therefore, the large differences in the phosphorylation of Erk1/2 (Wret-

man et al., 2001) and in its downstream kinases p70^{S6K} at Ser⁴²⁴/Thr⁴²¹ (Deldicque et al., 2008) and rpS6 at Ser^{235/236} (Pende et al., 2004; Roux et al., 2007), and possibly also in the phosphorylation of p38 γ , between the present exercise protocols soon after exercise, may be partly related to the larger metabolic/ionic perturbations during the hypertrophic protocol. For the p-Erk1/2, alternatively, the reason why no elevation was found with the 1 RM protocol may have been also partially due to timing of the biopsies (Tannerstedt Apro & Blomstrand, 2009), or just that the exercise did not induce enough long-lasting time-tension integral during each set (Martineau & Gardiner, 2001).

What could be the consequences of the observed differences between the present exercise protocols?

An enhanced mTOR and MAPK signaling is positive for muscle size (Baar & Esser, 1999; Haddad &

Adams, 2004; Terzis et al., 2008; Shi et al., 2009; Ruvinsky et al., 2009) and protein synthesis response to a bout of resistance exercise (Fluckey et al., 2006; Kumar et al., 2009). It is thus possible that the observed increased MAPK signaling (p-Erk1/2, p-p38 γ , p70^{S6K} at Thr⁴²¹/Ser⁴²⁴ and rpS6 at Ser^{235/236}), and, to a smaller extent, also a trend in signaling downstream of mTOR (p70^{S6K} at Thr³⁸⁹, rpS6 at Ser^{240/244}) (Pende et al., 2004; Roux et al., 2007) after the hypertrophic protocol when compared with the maximal strength protocol, may be mechanisms contributing to larger expected changes in muscle hypertrophy after this hypertrophic type of a training mode (Fry, 2004). However, at least in previously untrained subjects, the difference between the less extreme type of maximal strength (~ 3 – 5 RM) and hypertrophic (~ 10 RM) exercises in the muscle protein synthesis response after a single resistance exercise (Kumar et al., 2009) or in muscle hypertrophy after months of training (Weiss, Coney & Clark, 2000; Campos et al., 2002; Wernbom, Augustsson & Thomee, 2007), may be small. The pure maximal 1 RM load training increases muscle strength (Atha, 1981) and to some extent even muscle size in previously untrained subjects (Sale et al., 1985) but clearly more acute and long-term research is needed. For instance, the effects of the present type of pure 1 RM protocols on acute signaling responses have not been investigated previously.

Importantly, MAPK signaling may not only regulate muscle hypertrophy adaptation, but the elevated phosphorylation of Erk1/2 may be involved in endurance training-like adaptations (Tesch, 1988; Nader & Esser, 2001; Higginson et al., 2002; Zhou et al., 2007). This may also be the case with p38 γ (Pogozelski et al., 2009). It can be speculated that at least the clear increase in Erk1/2 phosphorylation observed only after the hypertrophic resistance exercise protocol and parallel larger increases in its downstream signaling (p-p70^{S6K} at Ser⁴²⁴/Thr⁴²¹ and p-rpS6 at Ser^{235/236}) may lead to previously observed increases in expression of angiogenic factors (Gavin et al., 2007). Moreover, in the long term, this may be one mechanism leading to an increase in capillaries per fiber (maintained capillary density) compared with the 1 RM training (Tesch, 1988). This would probably, eventually, increase the local muscle high-intensity endurance capacity when compared with the maximal strength type of training, at least in previously untrained subjects (Campos et al., 2002).

Study limitations and suggestions for the future studies

Muscle biopsies were obtained 0.5 h after exercise, and while this is a good time-point for mostly fast signaling events, it may not be representative

of all signaling responses throughout the pathways presented in the present study. For instance, the phosphorylation of mTOR itself after exercise or differences in signaling downstream of mTOR between the exercise protocols may have turned out to be significant at another time-point. Also, the length of the 15×1 RM exercise was obviously longer and hence it is possibly that the biopsy time-point in that exercise was not totally comparable to the 5×10 RM although in both it was 0.5 h post-exercise. In the future, possible temporal differences between these exercise protocols should be investigated, as well as potential differences in the protein synthesis of different muscle fractions (e.g., myofibrillar, sarcoplasmic, nuclear and mitochondrial).

In conclusion both high-intensity, low-volume (maximal strength) and medium-intensity, high-volume (hypertrophy) exercises rapidly increase MAPK signaling and also to some extent signaling downstream of mTOR in the normal fed state. However, muscle MAPK signaling, especially of Erk1/2, is greater early after the hypertrophic than maximal strength exercise protocol in young adult males. This may mediate adaptations specific to these different types of training regimens in terms of both muscle growth and muscular endurance capacity.

Perspectives

Higher MAPK and, to smaller extent, also probably signaling downstream of mTOR after the hypertrophic protocol may explain the specific differences in muscle hypertrophy (Fry, 2004) and muscle endurance capacity (Campos et al., 2002) after different exercises. Thus, for general health and well being, the hypertrophic protocol may be preferable for the general population. However, we showed for the first time that even the pure maximal strength protocol induced increases in signaling downstream of mTOR and MAPK in previously untrained subjects. Indeed, contraction intensity appears to be a strong stimulus for mTOR signaling (Eliasson et al., 2006; Holm et al., 2010) and muscle protein synthesis (Holm et al., 2010; Kumar et al., 2009). The high muscle tension stimulus is unfamiliar to previously untrained individuals, and may be the reason why we observed increased signaling even after maximal strength exercise. It must be remembered that these responses may not occur in trained individuals because of reduced adaptability over time (Häkkinen et al., 1987). Therefore, future studies should compare these exercises also in subjects with years of training.

Key words: S6K1, mTOR, MAPK, skeletal muscle, hypertrophy.

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